Heat Check: Quantifying the Hot Hand in the NBA

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Introduction

In basketball culture, “heat checks” are defined as difficult shots taken by a player who has previously made several shots and “checks” whether they have that elite shooting efficiency. For our project, we classified heat checks as shots taken at least 15 ft or further away after having made two consecutive field goals.

Some of the sport’s greatest moments emerges from such heat checks and elite shooting. Thus, we want to test whether these heat checks provide an actual advantage or negatively affect the team. Ultimately, we want to see whether a players’ high confidence increases their ability to shoot the ball.

General Approach

What is a Hot Streak? When a player makes several consecutive shots, players and fans believe the next shot is more likely to go in — the "hot hand" belief. We ask: is this real, or a statistical illusion?

Our Heat Check Definition

- Shot taken over 15 ft from basket
- Previous two shots both made (prev_1_make, prev_2_make = 1)
- Both makes within the same game and within a 3-minute window

Why It Matters

- Does player confidence after a streak lead to riskier shot selection?
- Should defenses tighten coverage after consecutive makes?
- Does heat improve efficiency, or just volume?

Dataset

Source: NBA Stats API

Scope: Top 15 scorers per season, 2019–20 through 2023–24 (~100K+ shots)

Key Features: Shot location (LOC_X, LOC_Y), shot distance, game clock which is game_seconds_elapsed, GAME_ID / PLAYER_ID for streak detection, SHOT_MADE_FLAG as outcome

Why this source?

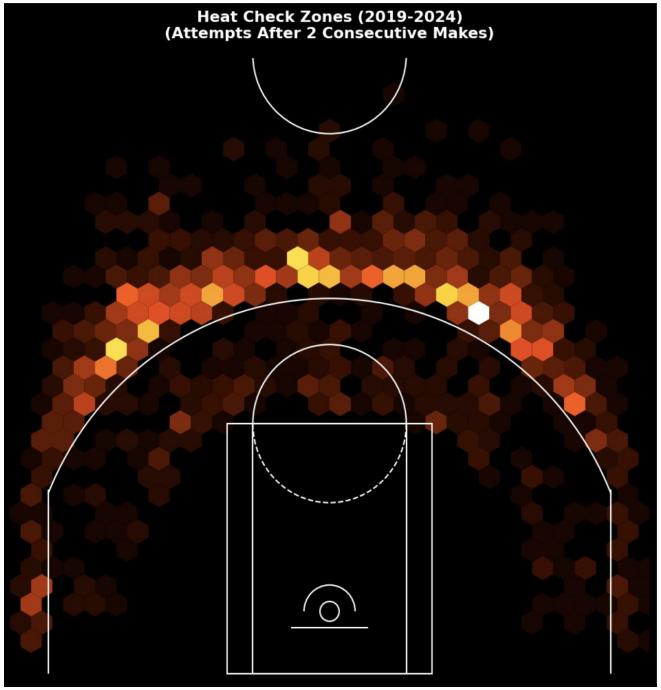
This API provides official NBA data with exact court coordinates. These are essential for spatial analysis. No other free source offers shot-level XY data.

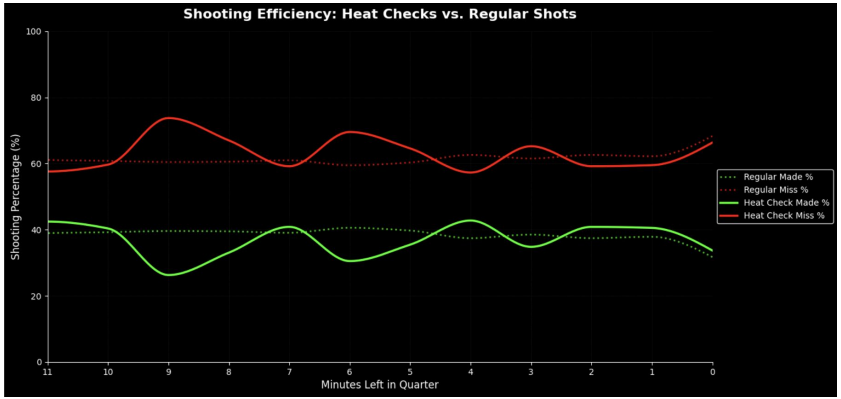
Limitations: Top-scorer bias, no defensive contest data, no fatigue/injury info, 2019-20 bubble season may be an outlier

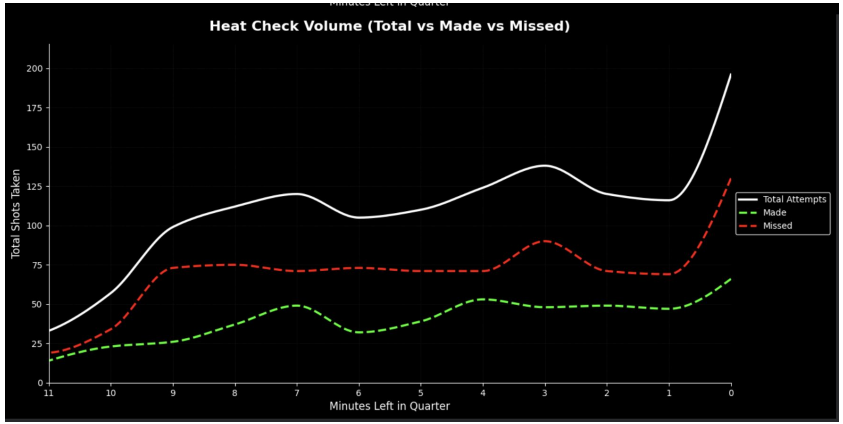
Data Collection

We collected NBA shot data by combining NBA stats API and nbaoncourt datasets. NBA stats API provides detailed play-by-play logs, including shot outcomes, timestamps, and game context, while nbaoncourt offers precise court location coordinates for each shot. By merging these datasets, we created a rich dataset capturing both temporal patterns (like previous makes) and spatial context (shot distance and angle). This combination allowed us to engineer features essential for predicting shot success and analyzing hot streaks across games and players.

Data Visualization







Feature Engineering

We transformed raw shot data into meaningful features to help the model predict shot success. Key features include:

- **Recent performance indicators** (prev_1_make, prev_2_make) to capture the context of a hot streak
- **Shot context** (SHOT_DISTANCE, LOC_X, LOC_Y, PERIOD, game_seconds_elapsed) to reflect difficulty of a shot and the relative time remaining when the shot was taken
- **Categorical feature for hot streaks** (is_heat_check). These engineered features combine player performance history and situational context, removing some of the confounding in the dataset.

Modeling

To train our model, we tried implementing two separate ensemble learning algorithms in our model, **XGBoost** and **Random Forest**, ultimately choosing **XGBoost**. Additionally, we implemented our own techniques to improve the model:

- **SMOTE:** Since the dataset is skewed with more misses, we generated synthetic examples of makes so the model actually learns the patterns of predicting makes instead of only predicting misses
- **Feature Importance:** Tweaking the importance of parameters listed in the feature engineering revealed that recent performance indicators (prev_1_make, prev_2_make) was the strongest predictor of the player's next shot, with the is_heat_check parameter being the third strongest predictor.

Conclusion

From our heat check analysis and model, we concluded that a player's momentum matters most when predicting the accuracy of their next shot, thus proving that the "heat check" is real to an extent. However, this also proves the unquantifiable variance in basketball as much of it relies on human elements that aren't quantifiable and by chance

What would help us more accurately predict the shot probability of a player is having defensive data that concurrently aligns with our datasets as the model could, more accurately, differentiate the validity and difficulty of each shot. However, our workarounds, such as shot distance, minutes left in quarter, and quarter proved to be ample substitutes for this issue